

Performance-Based Framework for Evaluating Truck Collision Risk for Bridge Piers

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Abstract: A risk-based method implemented within a performance-based design framework is proposed for protecting bridge piers subject to truck impact. Monte Carlo simulations are performed to calculate the probability of damage exceedance of impacted piers based on demand-to-capacity ratios. The demand is computed from the base shear of the pier under a calibrated pulse loading, which represents impact by a single-unit truck or a tractor-semitrailer. Four different highway classes are considered in this study with different ratios of truck types in the traffic volume, including single-unit trucks and tractor-semitrailer. Based on the simulation results, the probabilities of exceedance for different damage levels of typical or critical bridges, given a collision event, are provided for each highway class as a function of the posted speed limit and shear capacity of the impacted pier. The proposed methodology is verified through high fidelity simulations of truck collision with a full-bridge model and formatted in a manner that facilitates its direct adoption into current specifications. **DOI: 10.1061/(ASCE)BE.1943-5592.0001618.** © 2020 American Society of Civil Engineers.

Introduction

Highway bridge piers are usually at the risk of being impacted by trucks (Agrawal et al. 2018; Cao et al. 2019a; Deng et al. 2016). The current AASHTO (2017) specification prescribes two solutions to address vehicular impact with bridge piers: (i) providing structural resistance to an equivalent static force of 2,700 kN applied in a horizontal plane at a distance of 1,500 mm above the ground level; and (ii) redirecting or absorbing the collision load by protection components, such as a 1,370-mm-high TL-5 barrier located within 3,050 mm from the pier or a 1,100-mm-high TL-5 barrier located more than 3,050 mm from the pier.

To determine if pier protection is required, the current AASHTO (2017) recommends using the annual frequency for a bridge pier to be hit by heavy trucks (AF_{HBP}) to quantify the risk of truck impact with bridge piers. If AF_{HBP} is larger than 0.0001 for critical bridges or 0.0010 for other bridges, design for vehicular collision is required. AF_{HBP} is calculated based on the annual average daily truck traffic (AADTT) and the annual probability for a bridge pier to be hit by a heavy vehicle. However, the risk calculation in the current AASHTO (2017) does not consider the specific conditions of the bridge site and traffic on different highways (Ray et al. 2018). For example, no distinction is made between tangent sections and horizontally curved sections for undivided roadways, and the percentage of different truck types [i.e., single-unit versus tractor-semitrailer (TS)] in AADTT for different highways is not considered in the risk calculation in AASHTO (2017).

A recent study by Ray et al. (2018) proposed risk-based guidelines to determine whether pier protection is required for vehicular impact. In their study, details beyond those required in AASHTO (2017) for the site and traffic conditions were considered, including lane width, pier size, truck speeds, and traffic mix (ratio of different truck types among the total truck traffic volume). They calculated the annual frequency of bridge collapse based on the estimated annual number of heavy-vehicle encroachments and the probability of the estimated collision force exceeding the lateral capacity of the pier. Bridge collapse events were determined by comparing the equivalent static force and the nominal lateral resistance of the pier, where the static force was estimated based on the weight of the engine, the velocity of the truck, and the weight of the front parts of the truck body. Based on their criteria, pier failure is driven by engine impact. However, it is well known that truck loading consists of three impact events: bumper, engine, and trailer impact (Cao et al. 2019a). Although the engine impact usually causes the highest transient impact force, most of the weight and momentum of the truck comes from the trailer, which could lead to a complete collapse of the pier (Cao et al. 2019a). In other words, both engine impact force and trailer impact force would affect the damage levels of the pier and should be considered in determining the risk of pier failure by a heavy truck impact event. This is a key limitation of the Ray et al. (2018) study. Another limitation is that their design guidelines only considered binary pier performance, i.e., collapse or no collapse, instead of a more refined performance-based approach that specifies clear performance outcomes for given impact scenarios.

Recently, Agrawal et al. (2018) and Cao et al. (2019b) proposed a performance-based approach based on an extensive experimental and computational investigation on the design of a bridge pier subject to impact by a tractor-semitrailer (TS) weighing up to 360 kN or a single-unit (SU) truck weighing up to 118 kN. Through well-calibrated numerical simulation of TS and SU truck impacts on piers, impact force–time histories were proposed in the form of three triangular pulses corresponding to the three main impact events: bumper, engine, and cargo. The parameters of these functions were derived through numerical regression based on extensive simulation results. A performance-based approach that relates demands (in terms of the applied force–time histories) and capacity [in terms of acceptable shear distortion (SD) and plastic rotation (PR)] was developed for the design of bridge piers

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vulnerable to vehicular impact. However, this approach did not include a risk-based approach to decide the need for considering vehicular impacts during the design.

The objective of the current study is to address the shortcomings aforementioned by combining the approaches in Ray et al. (2018) and Agrawal et al. (2018) to develop new risk-based design guidelines for vehicle–pier impact situations within a performance-based design (PBD) framework. The proposed methodology is verified through high fidelity simulation and formatted in a manner that facilitates its direct adoption into current design specifications.

Stochastic Model of Truck Speed and Truck Type

This study considered the single-unit (SU) truck and the tractor-semitrailer (TS) as two types of large trucks that could potentially collide with the piers and cause damage or pier failure. As per the National Highway Traffic Safety Administration (NHTSA) report on Traffic Safety Facts, a large truck is defined as any medium or heavy truck, excluding buses and motor homes, with a gross vehicle weight rating (GVWR) greater than 45 kN (NHTSA 2014). These trucks will include both SU (GVWR > 45 kN but less than 116 kN) and TS (GVWR > 116 kN) trucks investigated in this paper. The NHTSA report indicates that 85% of the large trucks involved in fatal traffic crashes in 2014 were large, heavy trucks (GVWR > 116 kN, i.e., TS). This also implies that SU trucks were probably involved in the remaining 15% of the fatal traffic crashes. These data may not include all accidents involving impacts with bridge piers. However, the data do point to the two types of

trucks that pose a risk of impact to bridge piers. Hence, both truck types are studied in this paper to propose a generic risk-based method suitable for designing bridge piers against truck impact.

Fig. 1 shows the general shape of the three-triangle pulse model used to represent the force–time histories imposed by SU and TS trucks in frontal collisions with bridge piers (Agrawal et al. 2018; Xu et al. 2019; Cao et al. 2019c). For both truck types, the first peak represents the bumper impact. The second pertains to engine impact, while the third peak represents the cargo reaching the pier and colliding with it while compressing the engine between the pier and the cargo. Detailed illustration of the application of the proposed triangular pulse model for SU trucks and TS models can be found in Xu et al. (2019) and Cao et al. (2019c). The pulse models for both truck types are characterized by a total of 11 parameters. Parameters F_1 to F_5 correspond to the magnitude of peak forces, parameters T_1 to T_5 correspond to the time of application of the forces F_1 to F_5 , respectively, and T_6 represents the total duration of the pulse model.

The 11 parameters were estimated through nonlinear regression of impact force–time histories representing various impact scenarios (i.e., different combinations of truck weights, velocities, and pier sizes). Regression equations for pulse parameters F_1 to F_5 and T_1 to T_6 follow the general form of Eq. (1), where Y can represent any of the forces F_1 to F_5 in Newton and application times T_1 to T_6 in seconds; V is the truck impact velocity (kph); W is the truck weight (kN); and b is the pier width (mm). The regression parameters (α , β , γ , ϵ) are different for the SU truck and the TS model and can be found in Xu et al. (2019) and Cao et al. (2019c). These parameters are presented

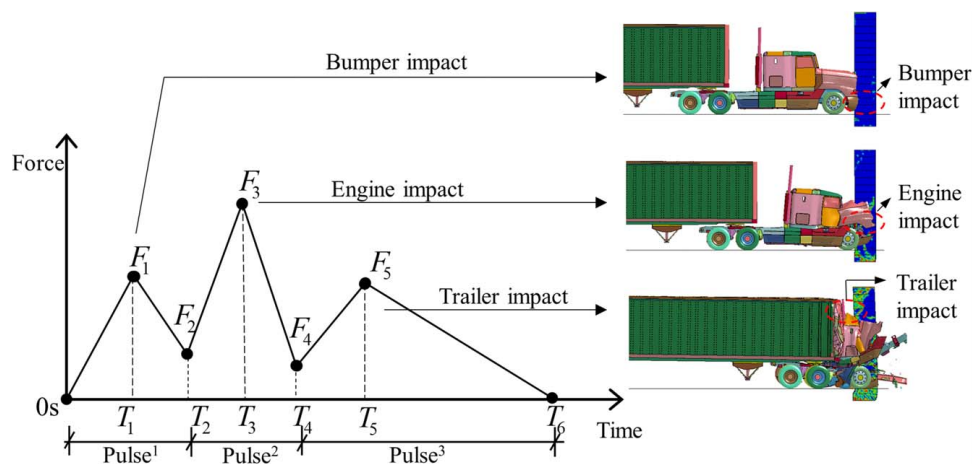


Fig. 1. The general shape of the pulse function.

Table 1. Equations of pulse parameters for SU truck and TS

Pulse parameters	SU truck				TS			
	α	β	γ	ϵ	α	β	γ	ϵ
F_1 (N)	2.32×10^5	0.58	0.00	0.71	3.78×10^5	0.52	0.00	0.33
F_2 (N)	2.41×10^4	0.88	0.00	2.38	1.81×10^5	0.51	0.00	−0.84
F_3 (N)	2.16×10^3	1.72	0.00	0.00	8.37×10^4	0.95	0.00	0.00
F_4 (N)	3.20×10^4	0.84	0.00	−0.12	8.14×10^3	1.08	0.00	−0.75
F_5 (N)	0.85	1.84	1.50	0.00	25.37	1.77	0.61	1.14
T_1 (s)	0.44	−0.89	0.00	0.00	0.41	−0.86	0.00	0.00
T_2 (s)	0.92	−0.90	0.00	0.00	2.43	−1.09	0.00	0.64
T_3 (s)	19.50	−1.46	0.00	0.00	1.62	−0.93	0.00	0.00
T_4 (s)	10.46	−1.25	0.00	0.00	4.76	−1.05	0.00	−1.21
T_5 (s)	0.09	0.00	0.00	0.00	0.22	0.00	0.00	0.00
T_6 (s)	0.17	0.00	0.00	0.00	0.80	0.00	0.00	0.00

in Table 1 for the sake of reference for both truck types. In reality, the information on truck velocity and truck weight could be obtained through weigh-in-motion technologies. Recently, He et al. (2017, 2019) and Deng et al. (2018) proposed a series of innovative methods, including the virtual simply supported beam method, equivalent shear force method, and virtual axle method, for detecting the speed and axle weight of vehicles passing over bridges without the need for installing axle detectors on the road. These methods are very convenient to use and have great potential for application

$$Y = \alpha(V)^{\beta}(W)^{\gamma}\left(\frac{b}{900}\right)^{\epsilon} \quad (1)$$

To consider the randomness of the impact demand, both truck speed and truck types were considered as random variables, which were used to generate the demand model shown in Eq. (1) and Fig. 1. For the truck speed, a normal distribution was assumed based on the study by Ray et al. (2018), where the mean speed of the truck was assumed to be 96% of the posted speed limit (PSL) in the highway, and the standard deviation of the truck speed was 5% of the PSL. The truck type (SU or TS) represents different levels of hazards to the highway bridges and the mix ratio of the two truck types can vary widely from one functional highway class to the other. The mix ratios of the SU truck and TS for different highways are shown in Table 2, which is based on the traffic data provided by Ray et al. (2018). In Table 2, it is assumed that buses will impose the same crash demands as single-unit trucks and are lumped under the SU truck category. Similarly, single-trailer and multi-trailer trucks are lumped under the TS truck category. These assumptions can be refined in the future as more data about the impact characteristics of buses and multitrailer trucks becomes available. Passenger cars (sedans and SUVs) are not considered in this study because they generally have much lower kinetic energy than the SU and TS categories. It can be seen from Table 2 that rural highways generally have a larger ratio of TS than urban highways, and interstate highways generally have a larger ratio of TS than collectors. To consider the randomness of truck type, a uniform random variable was used to determine the number of samples for SU and TS trucks based on the mix ratios shown in Table 2.

To generate the pulse demand using Eq. (1), the weight of TS was assumed to be 360 kN, and the SU truck was assumed to weigh 78 kN. The width of the impacted pier was assumed to be 900 mm. The generated pulse model was applied to an elastic pier model with the top and bottom fully constrained and time-history analysis was conducted to compute the base shear, D . Sufficient fixity generated from the top beam has been observed in several truck impact accidents documented by Buth et al. (2010). Also, the fixed-fixed boundary condition made the model more tractable and provided better control over the design variables. Detailed justifications for using the fixed-fixed boundary condition for the pier can be found in Agrawal et al. (2018), Xu et al. (2019), and Cao et al. (2019a, c).

Table 2. Mix ratio of truck types

Vehicle category	Rural		Urban	
	Interstates and arterials	Collectors and locals	Interstates and arterials	Collectors and locals
SU trucks	0.25	0.70	0.35	0.80
TS trucks	0.75	0.30	0.65	0.20

Risk Calculation

In the studies by Agrawal et al. (2018) and Cao et al. (2019b), three different performance levels for a pier subject to truck impact were proposed, i.e., immediate use (minor damage), damage control (moderate damage), and near collapse (severe damage). The three performance levels are defined as follows:

1. Immediate use: The bridge pier suffers minor damage, and the bridge is fully functional immediately after the impact. The failure modes associated with the minor damage state are tensile cracking of concrete and minor yielding of the longitudinal reinforcement. Both shear distortion (SD) and plastic rotation (PR) in this case should be less than 0.010 rad.
2. Damage control: The bridge pier suffers moderate damage in an impact event. The bridge pier will need to be repaired, which may require the closure of the bridge. Moderate damage is defined by the occurrence of major yielding and plastic deformations within the steel reinforcement as well as minor concrete core deterioration. Shear distortion and plastic rotation range from 0.010 to 0.075 rad.
3. Near collapse: The bridge pier suffers severe damage, but the bridge will not collapse. It may be cost-prohibitive to repair the bridge, or the pier may need to be replaced after the impact event. Severe damage is associated with plastic hinge formation in the pier and/or major damage to the concrete core. Shear distortion and plastic rotation in this case could range from 0.075 to 0.150 rad. The pier maintains sufficient integrity to prevent the truck from passing through. Since test data on the pier deformation under the truck impact are limited, the selected values of SD and PR for each performance level are based on the previous calibration work by Cao et al. (2019b) and can be refined in the future as more data become available.

The performance-based design process in Agrawal et al. (2018) and Cao et al. (2019b) first starts by applying the loading time history, as specified in Eq. (1), to a structural model of the pier. Like the applied loading history, the base shear demand also has three peaks (N), $D1$, $D2$, and $D3$. The computed demands are compared to the capacity limits to determine structural performance. As discussed in Cao et al. (2019b), the method differentiates between the full shear capacity of the pier from both concrete and shear reinforcement, C (N), and the shear capacity attributed to just the shear reinforcement, C_s (N). Pier performance is determined by computing the ratios of demand to capacity as follows, where the key variables are defined in Tables 3 and 4. It is noted that the bumper impact force is generally smaller than the engine impact and would not typically cause severe damage to the pier. Hence, only engine impact demand ($D2$) and trailer impact demand ($D3$) are considered in Tables 3 and 4.

- Immediate use (minor damage): $A0 < D2/C < A1$ for engine impact and $B0 < D3/C < B1$ for cargo impact.
- Damage control (moderate damage): $A0 < D2/C < A1$ for engine impact and $B1 \leq D3/C < B2$ for cargo impact.

Table 3. Performance levels, corresponding damage state, shear distortion, or plastic rotation for truck impact

Performance level	Damage state	Max(SD, PR)	$D2/C$	$D3/C$	$D3/C_s$
Immediate use	Minor	[0.000, 0.010]	[A0, A1]	[B0, B1]	N/A
Damage control	Moderate	[0.010, 0.075]	[A0, A1] [A1, A2]	[B1, B2] N/A	N/A [B1, B2]
Near collapse	Severe	[0.075, 0.150]	[A1, A2]	N/A	[B2, B3]

Note: SD = shear distortion; PR = plastic rotation; and N/A = not applicable.

Table 4. *D/C* ratios for different performance levels for SU truck and TS

Impact event	<i>D/C</i>	SU truck	TS
Engine impact	A0	0.00	0.00
	A1	2.50	2.00
	A2	3.75	2.75
Cargo impact	B0	0.00	0.00
	B1	1.50	0.80
	B2	2.25	1.30
	B3	3.00	1.60

Table 5. *D/C* ratios for evaluating the damage exceedance levels of critical and typical bridges

Bridge type	Unfavorable damage states	Deformation level	<i>D2/C</i>	<i>D3/C</i>	<i>D3/Cs</i>
Critical	Moderate, severe, and collapse	Max(SD, PR)	[A0, A1]	[B1, B2]	N/A
		> 0.010	[A1, A2]	N/A	>B1
			>A2	N/A	N/A
Typical	Severe and collapse	Max(SD, PR)	[A1, A2]	N/A	>B2
		> 0.075	>A2	N/A	N/A

Note: SD = shear distortion; PR = plastic rotation; and N/A = not applicable.

- $A1 \leq D2/C < A2$ for engine impact and $B1 \leq D3/Cs < B2$ for cargo impact.
- Near collapse (severe damage): $A1 \leq D2/C < A2$ for engine impact and $B2 \leq D3/Cs < B3$ for cargo impact.

The specific values for A0, A1, A2, B0, B1, B2, and B3 are also listed in Table 4. These values are essentially the thresholds of *D/C* ratios corresponding to different performance levels.

In this study, a range of shear capacities (*C*) was selected, and the stochastic demand-to-capacity (*D/C*) ratios were computed for each shear capacity value. The *D/C* ratios were then used to estimate the performance of the pier, given a collision event by following the performance-based design (PBD) criteria discussed previously and presented in Agrawal et al. (2018) and Cao et al. (2019b).

Per AASHTO (2017), critical bridges are generally required to be open to all traffic or emergency vehicles after the impact event. Therefore, the probability of exceedance for moderate damage (including moderate, severe damage, and collapse), $P(\text{EMD}|\text{collision})$, was calculated in this study to determine if pier protection is required for critical bridges. For typical bridges, the probability of exceedance for severe damage (including severe damage and collapse), $P(\text{ESD}|\text{collision})$, was calculated for the risk assessment. Table 5 shows the range of *D/C* ratios used to evaluate the damage exceedance levels of critical and typical bridges based on the PBD criteria proposed by Agrawal et al. (2018) and Cao et al. (2019b). It should be noted that *D/C* ratios exceeding the severe-damage thresholds were considered to be related to pier collapse, which is also an unfavorable damage state for both critical and typical bridges. The specific values of A0 to A2 and B1 and B2 for both trucks are shown in Table 4.

A detailed flowchart of the risk calculation is shown in Fig. 2. As shown in Fig. 2, the probability of damage exceedance, given a collision event, was calculated as the number of instances where the damage to the pier exceeded certain levels (estimated by *D/C* ratios for moderate and severe damages for critical and ordinary bridges, respectively), divided by the total number of the demand samples. Based on the flowchart in Fig. 2, Monte Carlo simulations were carried out by using $N=25,000$ cases per

category for speeds ranging from 64 to 113 kph in increments of 16 kph for the four highway classes shown in Table 2.

In Fig. 2, the shear capacity of the pier was increased from 0 to 5,500 kN in increments of 220 kN. For a specific speed, the generated 25,000 stochastic demand samples were compared with each of the shear capacity values (i.e., the *D/C* ratio was computed) and were used to assess the performance level of the pier. The ratio of the number of cases that exceeded the selected performance criterion to the total number of cases was then used to compute the probability of damage exceedance.

Simulation Results of Probability of Damage Exceedance

Based on the Monte Carlo simulation, Figs. 3 and 4 show the cumulative distribution of the probability of damage exceedance for critical and typical bridges, given a collision event for the four different highways (i.e., rural interstate, rural collectors, urban interstate, and urban collectors). In Figs. 3 and 4, the probability of damage exceedance decreased from 1 to 0 as the shear capacity increased up to its maximum value. The plots also show that, for a certain shear capacity, the risk of pier collapse increased as the velocity increased. For each highway class, the risk curves showed a characteristic shape, where a plateau occurs at the risk level consistent with the ratio of TS trucks to the whole truck population (Table 2). For example, the plateau level was located at 0.75 for rural interstates shown in Fig. 3(a) and 0.30 for rural collectors, as shown in Fig. 3(b). In Table 2, the ratio of TS over the whole traffic volume is 0.75 for rural interstates and 0.30 for rural collectors. The plateau level represents a transition in bridge pier behavior as the type of truck changes from a mix of SU trucks and TS to predominantly TS. In particular, the latter generally causes a higher base shear demand than that by the SU truck and, therefore, most of the damage exceedance cases would be caused by TS as the shear capacity grows. The computed risk curves shown in Figs. 3 and 4 can be used by engineers to determine if pier protection is required for critical and typical bridges based on the posted speed limit (PSL) and shear capacity of the pier. In the Appendix, the computed probability values shown in Figs. 3 and 4 are tabulated based on different shear capacities and PSL, which facilitate its direct adoption into current specifications.

Verification of the Risk Values by Finite-Element Simulations

The Monte Carlo simulations showed that the probability of damage exceedance for moderate and severe damage is closely related to the travel speed of the truck and the shear capacity of the bridge pier. High fidelity finite-element (FE) simulations of truck impact were conducted to show that this trend is meaningful and provide confidence in the obtained results. The simulations, which were conducted using the modeling scheme in Cao et al. (2019b), involve a 360-kN TS model impacting a full-bridge model at impact speeds of 80 and 113 kph. The bridge was assumed to be rated as a critical one located on a rural-interstate highway system.

Bridge Model

Liu (2012) investigated the impact of an SU truck on the pier-bent of a three-span simply supported bridge through numerical simulation. The model, which was developed by Yi (2009) and Yi et al. (2013a, b), is shown in Fig. 5(a). Fig. 5(b) shows a close-up of

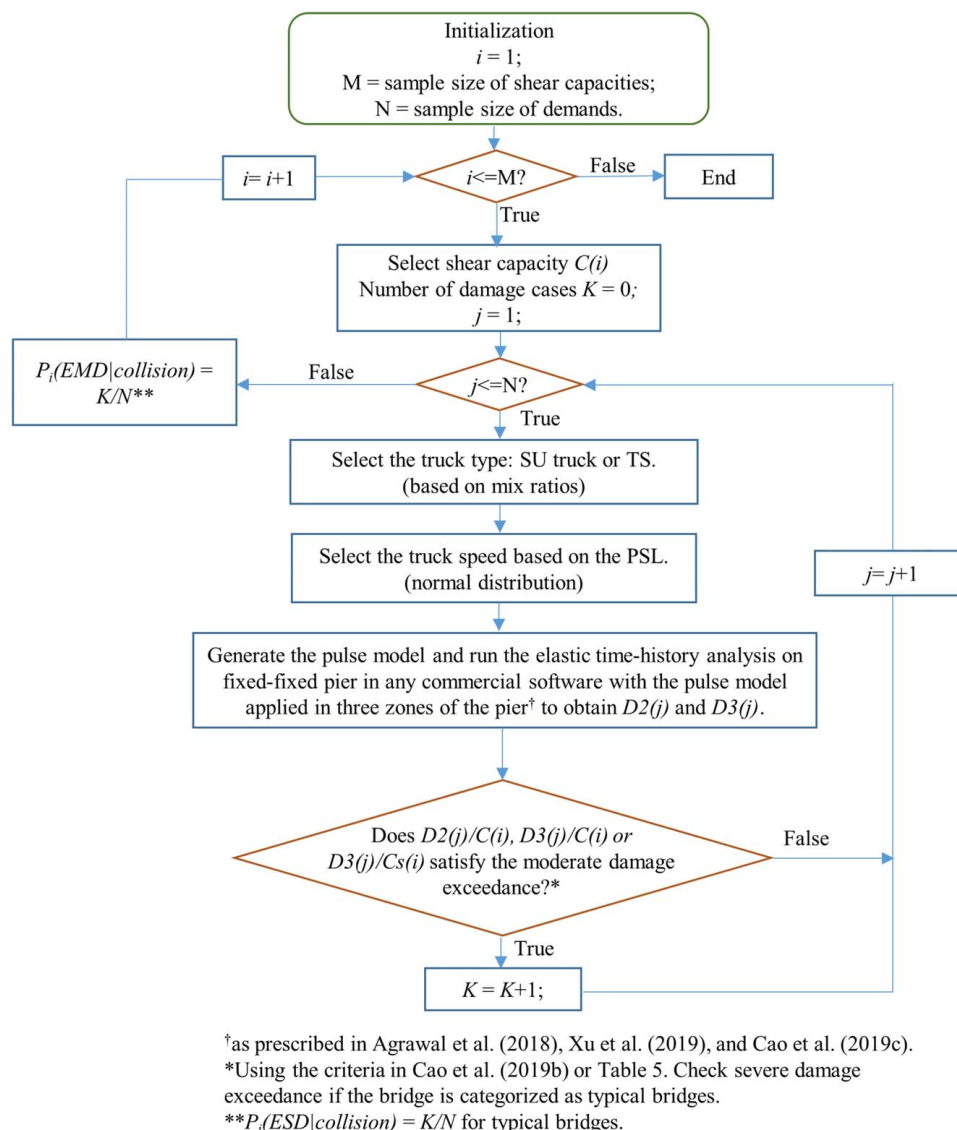


Fig. 2. Flowchart for computing $P(EMD|collision)$ for critical bridges for a specific posted speed limit (PSL).

the finite-element setup. In the FE model of the bridge, footing, pier, bent bearing, and deck are modeled with solid elements, stringers; diaphragms are modeled with shell elements; and steel reinforcements are modeled with beam elements. Concrete in the bridge model was modeled using reduced integration solid elements and concrete behavior was modeled using the Continuous Surface Cap Model (MAT 159) in LS-DYNA (Hallquist 2006). The model can capture key aspects of inelastic concrete behavior, including postpeak softening and confinement effects. It is based on an elastoplastic damage formulation that accounts for rate effects. Detailed input parameters for this concrete model can be found in Xu et al. (2019). Steel bars were represented using Hughes–Liu beam elements and were modeled using material MAT 3 in LS-DYNA (Hallquist 2006), which account for yielding and kinematic plasticity hardening. A perfect bond was enforced between steel and concrete components, implying that the effect of bond-slip behavior was assumed to be negligible.

The bridge pier had a clear height of 4,900 mm and a square section of 900 mm × 900 mm. The pier was proportioned to have a shear capacity of 3,300 kN based on the capacity design procedure outlined in Agrawal et al. (2018) and Cao et al. (2019c).

Truck Model

The TS model was initially developed by Miele et al. (2010) and is illustrated in Fig. 6. The truck model is a van-type tractor-semitrailer with dimensions of 19.4 m × 2.6 m × 4.0 m and is representative of models such as 1991 White GMC tractor with a 1988 Pines 14.6-m semitrailer, 1979 International TranStar 4200 tractor with a 1977 Pullman van-trailer, and 1992 Freightliner FLD120 with integral sleeper cabin or a 1990 Stoughton box semitrailer (Miele et al. 2010). The finite-element model of the truck is composed of approximately 472,000 finite elements, including shell, beam, and solid elements for different parts of the truck. Further detailed information about the model can be found in Miele et al. (2010) and Cao et al. (2019a, b, c).

Validation

The truck model has been validated against two full-scale barrier crash tests (Miele et al. 2010) and another head-on crash test involving a rigid pier (Cao et al. 2019a). Per Miele et al. (2010), the tractor-semitrailer vehicle model developed is the most advanced of this vehicle class in terms of physical function, geometric

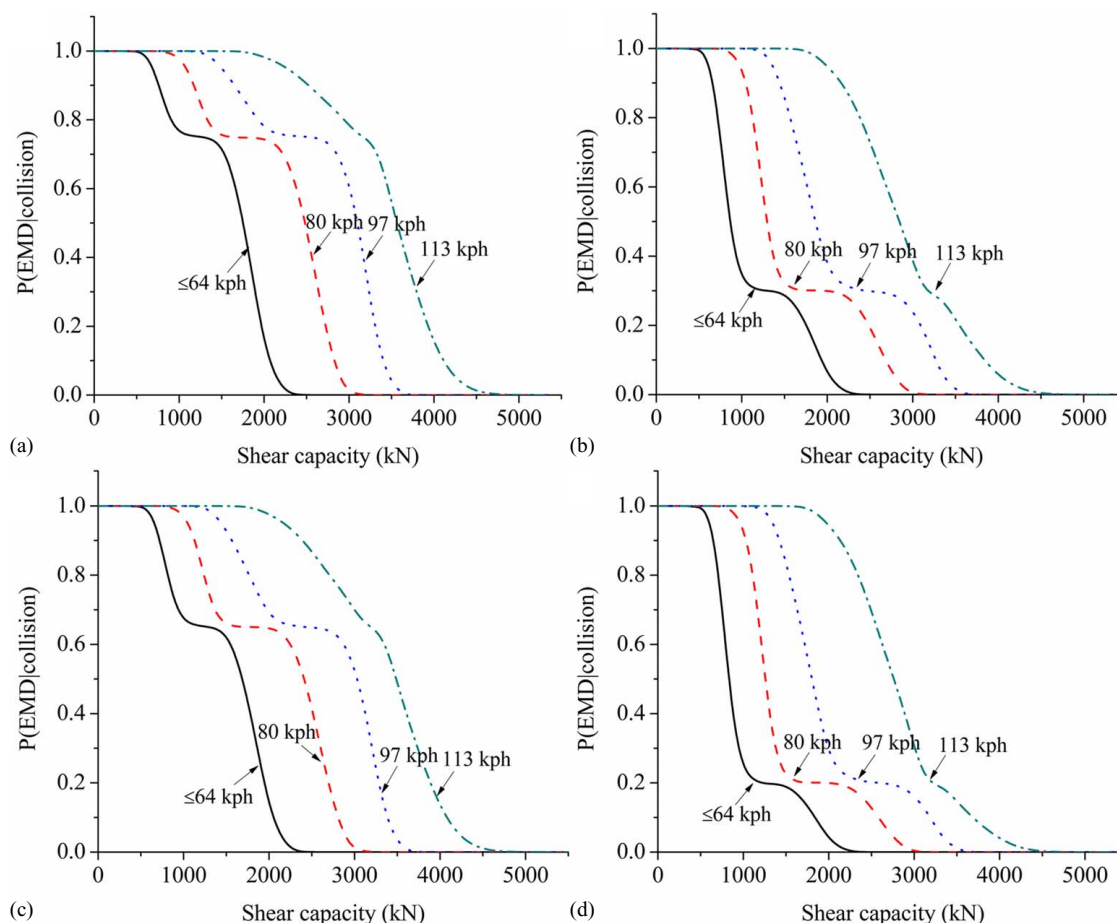


Fig. 3. Risk curves of $P(\text{EMD}|\text{collision})$ for critical bridges: (a) rural interstates and arterials; (b) rural collectors and locals; (c) urban interstates and arterials; and (d) urban collectors and locals.

detail, and material property accuracy. The model is computationally robust, given its complexity. It has been extensively debugged and exercised hundreds of times for the cases within the range of NCHRP Report 350 TL-5 speeds and impact angles. The selected modeling scheme with different truck speeds, weights, and detailing of the structural designs has yielded reasonable results in multiple publications including Agrawal et al. (2018), Xu et al. (2019), Cao et al. (2019a, b, c, 2020), and Chen et al. (2019). Therefore, the probabilistic variation in the vehicle, such as speed in this study, may not affect the formulation of the presented framework.

The reinforced concrete (RC) modeling scheme has also been widely validated by many tests from cylinder compression testing to static pushover testing on concrete barriers and impact cart testing (Murray et al. 2007). The variation of the concrete model, such as the strength of the concrete, is well considered in the model development and usage and has been proven to yield reasonable simulation results. Recently, the material model has been further calibrated against a series of drop hammer tests on RC beams with different reinforcement (Xu et al. 2019; Cao et al. 2019a, c). The simulation results showed that the selected RC modeling can capture both flexural and shear damage in the concrete beam under impact loading. The most recent validation of the RC modeling was conducted by Chen et al. (2019) based on two pendulum impact tests with a half-scale pier-bent model. In their simulation, the simulated damage and deformation of the pier-bent model under the pendulum impact matched the testing results reasonably well.

Simulation Results

Fig. 7(a) shows the setup of the truck impact with the full-bridge model in LS-DYNA (Hallquist 2006). The damage modes of the bridge piers by a 360-kN heavy truck impact at 80 and 113 kph are shown in Figs. 7(b and c), respectively. Based on the simulation results, the pier only suffered minor damage (with a maximum shear distortion around 0.003) by the 80-kph impact [Fig. 7(b)], but it was severely damaged (with a maximum shear distortion of around 0.150) by the 113-kph impact [Fig. 7(c)]. Also, the time histories of the displacement of the pier at the impact location (denoted in Fig. 7) are plotted in Fig. 8. In Fig. 8(a), the permanent deformation of the pier is low, roughly 3 mm, for the 80-kph impact, which is representative of minor damage. In Fig. 8(b), however, the residual deformation of the pier increased to 140 mm during the 113-kph impact, implying significant damage to the pier, as also observed from Fig. 7(c). It should be noted that the bridge did not collapse under the 113-kph impact.

Based on the risk curves shown in Fig. 3(a), $P(\text{EMD}|\text{collision})$ is 0.0000 for critical bridges with piers having a shear capacity of 3,300 kN when the speed of the truck is 80 kph, showing a low probability of exceedance for moderate damage. $P(\text{EMD}|\text{collision})$ increased to 0.7317 when the speed of the truck is 113 kph, showing a high probability of severe damage. Clearly, the damage modes shown in Fig. 7 match the risk predictions of $P(\text{EMD}|\text{collision})$ for 80-kph (minor damage) and 113-kph impact (severe damage). While not comprehensive, this exercise provides assurance

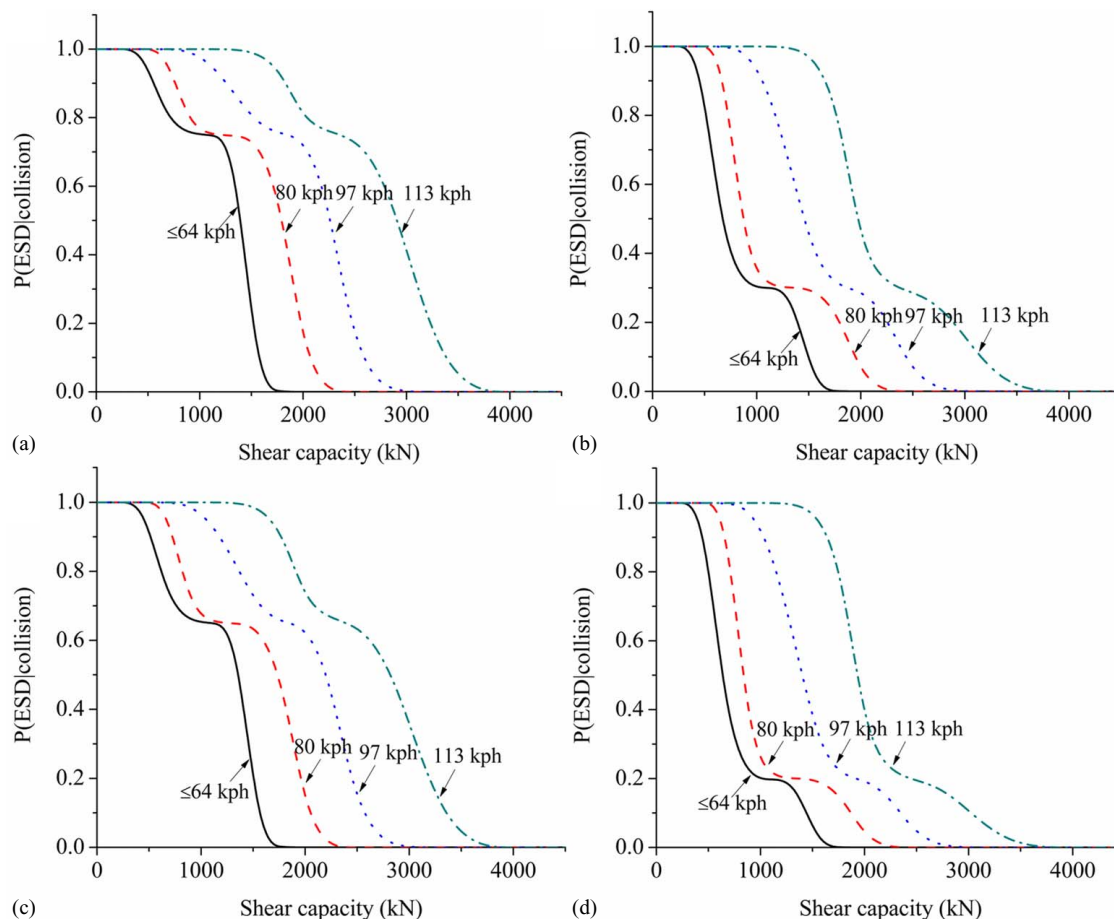


Fig. 4. Risk curves of $P(\text{ESD}|\text{collision})$ for typical bridges: (a) rural interstates and arterials; (b) rural collectors and locals; (c) urban interstates and arterials; and (d) urban collectors and locals.

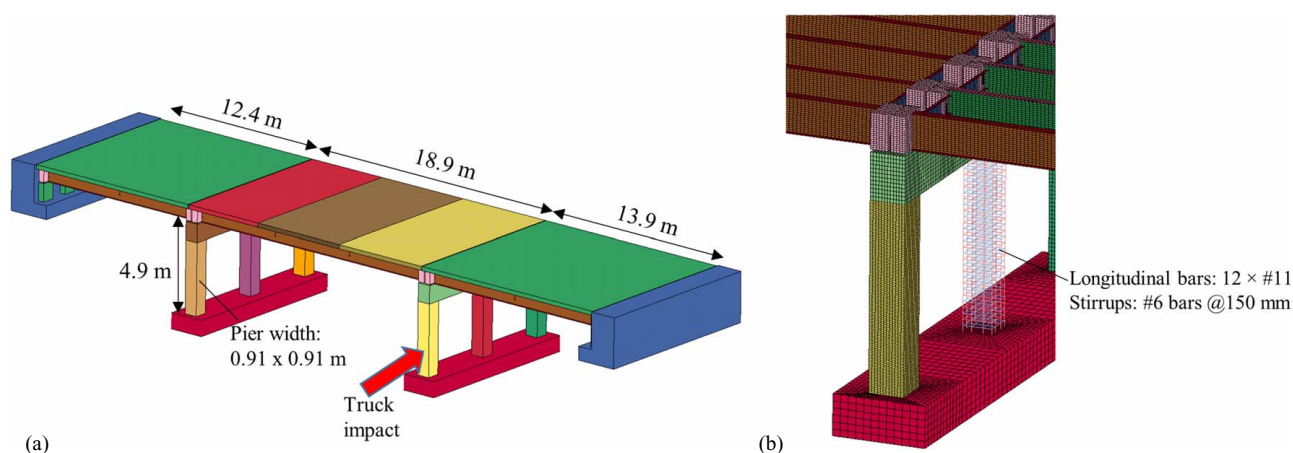


Fig. 5. FE model of the bridge: (a) whole bridge model; and (b) close-up of the bridge pier.

that the proposed risk methodology yields results that are representative of the actual risk of damage during an impact event.

Application of the Probability of Damage Exceedance

To determine whether pier protection against truck impact is required, the proposed probability of damage exceedance, given a collision event [i.e., $P(\text{ESD}|\text{collision})$ or $P(\text{EMD}|\text{collision})$], can

be further combined with the probability of truck collision based on different site and traffic conditions.

In the recent NCHRP report by Ray et al. (2018), the calculation of annual frequency of truck collision, AF_{TC} , was presented by

$$\text{AF}_{\text{TC}} = \sum_{i=1}^m N_i \cdot \text{HVE}_i \cdot P(\text{collision}|\text{HVE}_i) \quad (2)$$

where i = approach direction 1, 2, ..., m ; m = total number of approach directions; N_i = site-specific adjustment factor; and

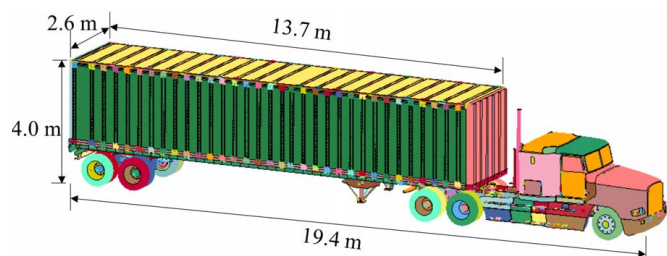


Fig. 6. Tractor-trailer model.

HVE_i = heavy-vehicle base encroachment frequency; and $P(\text{collision}|HV;E_i)$ is the probability of a collision, given a heavy-vehicle encroachment. These parameters are dependent on the traffic condition and bridge site configurations. The base encroachment frequency (HVE_i) was developed based on the so-called Cooper encroachment data using the risk analysis tool RSAPv3 (Ray et al. 2012) considering different highway types and annual average daily traffic (AADT). The site-specific adjustment factor, N_i , modifies the ideal base condition value (HVE_i) to account for the particular features of the site, such as lane width, vertical grade, and horizontal curvature. The third parameter $P(\text{collision}|HV;E_i)$ was

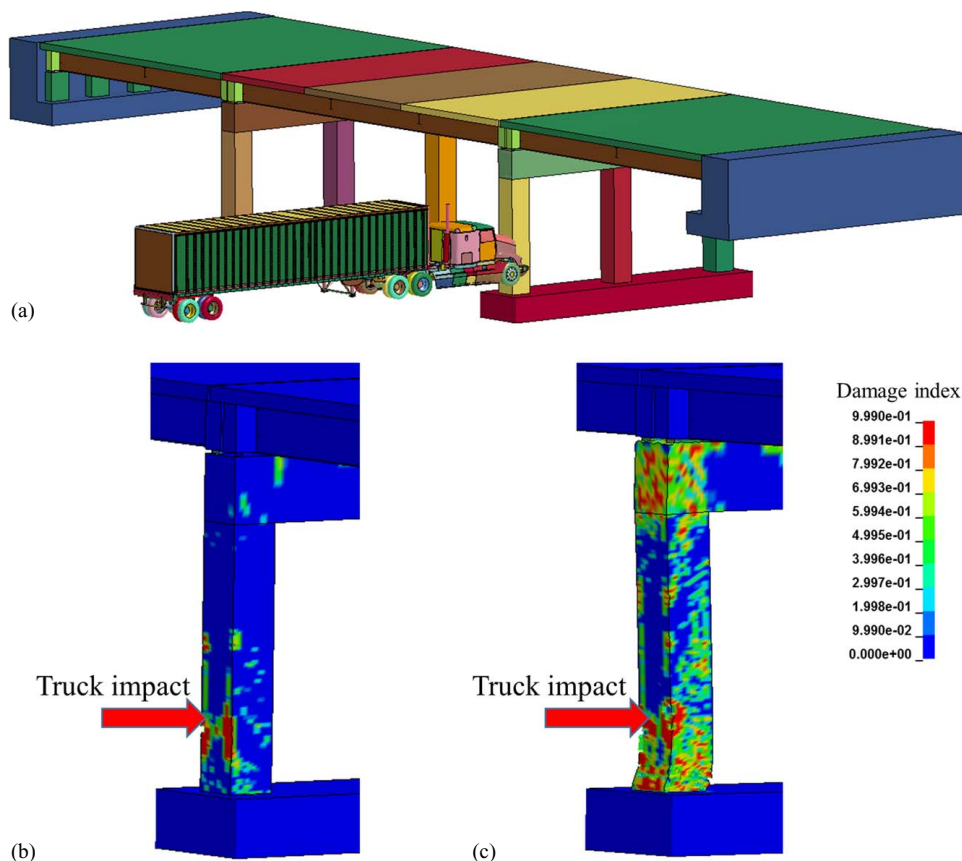


Fig. 7. Damage mode of the pier collided by the heavy truck: (a) general setup of the truck simulation in LS-DYNA; (b) 80 kph (minor damage); and (c) 113 kph (severe damage).

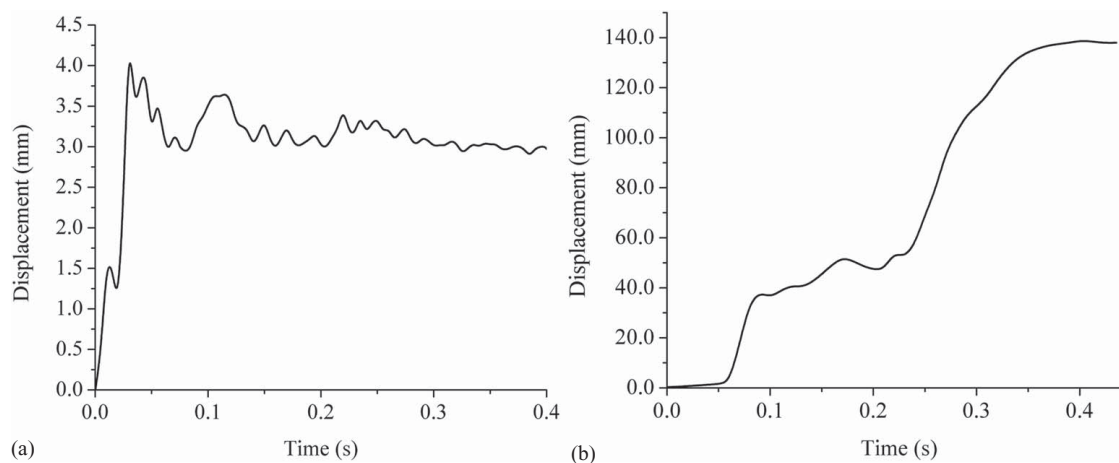


Fig. 8. Time history of displacement of the bridge pier: (a) 80 kph impact; and (b) 113 kph impact.

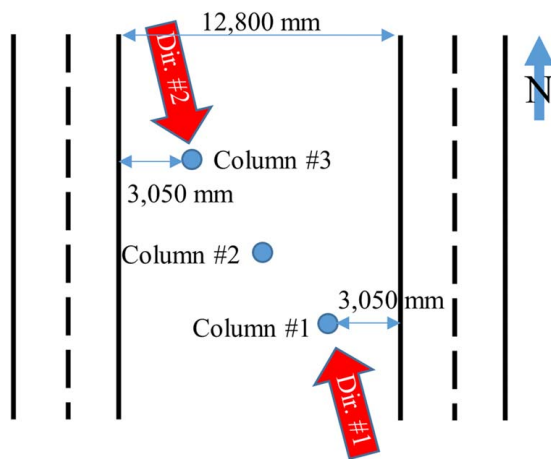


Fig. 9. Example site layout.

developed by Ray et al. (2018) based on a database of simulated heavy-vehicle trajectories that encroached onto the roadside within 90 m of the pier component using RSAPv3 (Ray et al. 2016). The predicted probability of a heavy-vehicle crash was dependent on the pier component offset and diameters of the pier. Details of these values can be found in Ray et al. (2018) and their calculations will be illustrated in the design example.

Based on Eq. (2), the annual frequency of bridge damage exceedance from truck impact, AF_{BDE} , can be calculated by combining AF_{TC} and $P(ESD|collision)$ or $P(EMD|collision)$ following Eqs. (3) and (4). Design for vehicular collision may not be required if AF_{BDE} is less than 0.0001 for critical or essential bridges, as defined in the AASHTO (2017) Article 1.3.5 or 0.0010 otherwise.

If AF_{BDE} is greater than or equal to 0.0001 for critical or essential bridges or 0.0010, otherwise bridge piers should be either designed to resist vehicular collision forces using the performance-based approach or shielding using barriers, as described in AASHTO (2017) Article 3.6.5.2.

The thresholds for AF_{BDE} (i.e., 0.0001 for critical bridges and 0.0010 for typical ones) mirror those for vessel collision force found in AASHTO (2017) Article 3.14.5

For critical bridges:

$$AF_{BDE} = \sum_{i=1}^m N_i \cdot HVE_i \cdot P(collision|HVE_i) \cdot P(EMD|collision) \quad (3)$$

For typical bridges:

$$AF_{BDE} = \sum_{i=1}^m N_i \cdot HVE_i \cdot P(collision|HVE_i) \cdot P(ESD|collision) \quad (4)$$

Risk Calculation Example

An example bridge site was used to illustrate the method of risk calculation prescribed in Eqs. (3) and (4) for typical and critical bridges. As shown in Fig. 9, the bridge was assumed to be located in a four-lane divided rural interstate. The example was based on the verification work presented in the NCHRP report by Ray et al. (2018).

Bridge Site Information

The layout for the bridge site is shown in Fig. 9, where it represents a three-pier system located in the center of a 12,800-mm-wide

Table 6. Bridge site information

Bridge/traffic characteristics	Value	
Shear capacity: C (kN)	3,300	
Pier size (mm)	910	
	Direction 1	Direction 2
Highway type	Divided	Divided
Functional classification	Rural interstate	Rural interstate
Two-way AADT (veh/day)	50,000	50,000
Percent trucks (PT) (%)	25	25
Offset to critical pier component (mm)	3,050	3,050
Major accesses (points)	1	1
Lanes in one direction	2	2
Lane width (mm)	3,650	3,650
PSL (kph)	113	113
Grade	+4%	−4%

Table 7. Site-specific adjustment factors

Adjustment factor	Direction 1	Direction 2
Major accesses (f_{ACC})	2.00	2.00
Lane width (f_{LW})	1.00	1.00
Horizontal curve radius (f_{HC})	1.00	1.00
Lanes in one direction (f_{LN})	1.00	1.00
Posted speed limit (f_{PSL})	1.00	1.00
Grade (f_G)	1.00	1.50
Site-specific adjustment factor (N_i)	2.00	3.00

Source: Reprinted from Ray et al. (2018).

median of a rural divided interstate with 50,000 vehicles/day, where 25% of the vehicles are trucks with a PSL of 113 kph. The three piers are placed on a skew such that the leading pier in each direction is 3,050 mm from the edge of the traveling lane. All three piers are 910 mm in diameter, and the shear capacity of each of the piers is calculated to be 3,300 kN. Key information about the bridge site is shown in Table 6.

Site-Specific Adjustment Factor: N_i

Based on the NCHRP report (Ray et al. 2018), the site-specific adjustment factor, N_i , can be calculated using Eq. (5). The key parameters for calculating N_i are listed in Table 7 based on the site conditions. Detailed development of these variables can be found in Ray et al. (2018). Per Table 7 and Eq. (5), N_i is calculated to be 2.00 for Direction 1 and 3.00 for Direction 2

$$N_i = f_{ACC} \cdot f_{LN} \cdot f_{LW} \cdot f_G \cdot f_{HC} \cdot f_{PSL} \quad (5)$$

Heavy-Vehicle Base Encroachment Frequency: HVE_i

Per Ray et al. (2018), the heavy-vehicle base encroachment frequency is correlated with annual average daily traffic (AADT) and the percentage of trucks among total vehicles (PT) on the highway. Table 8 shows the values of HVE_i for different PT and AADT provided by Ray et al. (2018). In this example, AADT = 50,000 and PT = 25%. Therefore, HVE_i is found to be 0.0065 for both directions per Table 8.

Probability of Collision Given a Heavy-Vehicle Encroachment: $P(collision|HVE_i)$

Per NCHRP report (Ray et al. 2018), the probability of an encroaching heavy vehicle striking the pier system, $P(collision|$

Table 8. Heavy-vehicle base encroachment frequency in direction i (HVE_i)

Two-way AADT vehicle/day	Divided highways							
	Percent trucks (PT)							
	5	10	15	20	25	30	35	≥40
1,000	0.0006	0.0006	0.0006	0.0006	0.0007	0.0007	0.0007	0.0007
5,000	0.0026	0.0026	0.0027	0.0027	0.0028	0.0028	0.0028	0.0028
10,000	0.0042	0.0043	0.0044	0.0045	0.0045	0.0045	0.0046	0.0046
15,000	0.0051	0.0053	0.0054	0.0054	0.0055	0.0055	0.0056	0.0056
20,000	0.0055	0.0057	0.0058	0.0059	0.0060	0.0060	0.0060	0.0061
24,000–47,000	0.0056	0.0058	0.0059	0.0060	0.0061	0.0061	0.0062	0.0062
50,000	0.0060	0.0062	0.0064	0.0065	0.0065	0.0066	0.0066	0.0067
55,000	0.0066	0.0069	0.0070	0.0071	0.0072	0.0072	0.0073	0.0073
60,000	0.0072	0.0075	0.0076	0.0077	0.0078	0.0079	0.0079	0.0080
65,000	0.0078	0.0081	0.0083	0.0084	0.0085	0.0085	0.0086	0.0087
70,000	0.0084	0.0087	0.0089	0.0090	0.0091	0.0092	0.0093	0.0093
75,000	0.0090	0.0094	0.0095	0.0097	0.0098	0.0099	0.0099	0.0100
80,000	0.0096	0.0100	0.0102	0.0103	0.0104	0.0105	0.0106	0.0107
85,000	0.0102	0.0106	0.0108	0.0110	0.0111	0.0112	0.0113	0.0113
≥ 90,000	0.0108	0.0112	0.0115	0.0116	0.0117	0.0118	0.0119	0.0120

Source: Reprinted from Ray et al. (2018).

Table 9. Probability of a heavy-vehicle collision, given a heavy-vehicle encroachment, $P(\text{collision}|HVE_i)$

Offset (mm)	Pier size (mm)				
	305	610	910	1,220	1,830
610	0.1763	0.1868	0.1978	0.2093	0.2337
1,219	0.1650	0.1750	0.1855	0.1964	0.2198
1,829	0.1543	0.1638	0.1738	0.1842	0.2064
2,438	0.1442	0.1532	0.1626	0.1725	0.1937
3,050	0.1347	0.1432	0.1521	0.1614	0.1816
4,572	0.1131	0.1204	0.1282	0.1363	0.1539
6,096	0.0946	0.1009	0.1075	0.1145	0.1297
7,620	0.0789	0.0842	0.0899	0.0958	0.1088
9,144	0.0656	0.0701	0.0749	0.0799	0.0910
10,668	0.0544	0.0582	0.0622	0.0665	0.0758
12,192	0.0450	0.0482	0.0515	0.0551	0.0630

Source: Reprinted from Ray et al. (2018).

HVE_i), is a function of the pier size and the distance between the face of the pier component and the edge of the travel lane, i.e., offset. Table 9 shows the values of $P(\text{collision}|HVE_i)$ for different pier sizes and offset. In this example, the pier size is 910 mm and the offset is 3,050 mm, as shown in Table 6. Hence, $P(\text{collision}|HVE_i)$ for each direction can be obtained from Table 9 as 0.1521.

Probability of Exceedance of Moderate Damage Given a Collision Event, $P(\text{EMD}|\text{collision})$

Given the shear capacity of the pier is 3,300 kN, for rural interstates with PSL = 113 kph, the probability of exceedance for moderate damage given a collision event, $P(\text{EMD}|\text{collision})$, was found to be 0.7317 from the Appendix.

Probability of Exceedance of Severe Damage Given a Collision Event, $P(\text{ESD}|\text{collision})$

Similar to $P(\text{EMD}|\text{collision})$, the probability of exceedance for severe damage given a collision event was found to be 0.1114 from the Appendix.

Table 10. Calculation results of annual frequency of damage exceedance for the example bridge site

Risk parameters	Direction 1	Direction 2
N_i	2.00	3.00
HVE_i	0.0065	0.0065
$P(\text{collision} HVE_i)$	0.1521	0.1521
$P(\text{EMD} \text{collision})$		0.7317
$P(\text{ESD} \text{collision})$		0.1114
$AF_{BDE}(\text{critical})$		0.0037
$AF_{BDE}(\text{typical})$		0.0006

Annual Frequency of Damage Exceedance for Critical Bridges, AF_{BDE}

Based on the calculations shown previously, the variables in Eq. (3) can be substituted to obtain the AF_{BDE} for critical bridges as follows:

$$\begin{aligned}
 AF_{BDE} &= \sum_{i=1}^m N_i \cdot HVE_i \cdot P(\text{collision}|HVE_i) \cdot P(\text{EMD}|\text{collision}) \\
 &= (2.00 \times 0.0065 \times 0.1521 + 3.00 \times 0.0065 \times 0.1521) \times 0.7317 \\
 &= (0.0020 + 0.0030) \times 0.7317 \\
 &= 0.0037 > 0.0001
 \end{aligned} \quad (6)$$

Annual Frequency of Damage Exceedance for Typical Bridges, AF_{BDE}

Similarly, the AF_{BDE} for typical bridges was calculated to be 0.0006 as follows:

$$\begin{aligned}
 AF_{BDE} &= \sum_{i=1}^m N_i \cdot HVE_i \cdot P(\text{collision}|HVE_i) \cdot P(\text{ESD}|\text{collision}) \\
 &= (2.00 \times 0.0065 \times 0.1521 + 3.00 \times 0.0065 \times 0.1521) \times 0.1114 \\
 &= (0.0020 + 0.0030) \times 0.1114 \\
 &= 0.0006 < 0.0010
 \end{aligned} \quad (7)$$

Summary

The results of risk calculations for this bridge site are summarized and shown in Table 10. Based on the values of AF_{BDE} , if the bridge is rated as a *critical* bridge, it requires shielding or should be designed for vehicular impact forces, because the probability of damage exceedance for moderate damage is larger than the threshold value (i.e., $0.0037 > 0.0001$). If the bridge is defined as a *typical* bridge, the pier system does not require shielding or specific design for the vehicular impact forces since the probability of damage exceedance for severe damage is less than the typical bridge threshold of 0.0010 (i.e., $0.0006 < 0.0010$).

Conclusion

In this study, a risk-based design approach for bridge piers subject to truck impact was developed based on a performance-based design (PBD) framework developed previously by the authors. Two random variables, truck speed and truck type, were used to generate stochastic pulse loading that represented the impact from single-unit trucks and tractor-semitrailers. The generated pulse loading was applied to a pier model to compute base shear demand. The base shear demand samples were collected and compared with a wide range of shear capacities of the pier to evaluate the performance of the pier during truck impact based on demand-to-capacity (D/C) ratios. For critical and typical bridges in four different highway classes, the number of damage exceedance instances was used to calculate the probability of the damage level exceedance of the pier in a given collision event. The proposed methodology was verified through high fidelity simulations of truck collision with a full-

bridge model and was formatted in a manner that facilitates its direct adoption into current specifications. The calculated probability of pier damage exceedance can serve as a simple but powerful tool for engineers to decide whether pier protection is required based on the highway conditions (posted speed limit and shear capacity of the pier).

The presented design methodology was calibrated to data that assumed a head-on truck impact. Although head-on crashes are considered to be the most severe impact scenarios, further studies could be conducted to generalize the results to situations involving noncircular sections subjected to an oblique impact, in which the approaching truck impacts the pier at an angle with respect to its primary axes.

Appendix. Exceedance Probability of Damage Levels for Typical and Critical Bridges Located in Rural and Urban Areas

Table 11 lists the computed probability values of $P(EMD|collision)$ for rural critical bridges shown in Figs. 3(a and b) based on different shear capacities and PSL. Table 12 lists the computed probability values of $P(ESD|collision)$ for rural typical bridges shown in Figs. 4(a and b) based on different shear capacities and PSL.

Table 13 lists the computed probability values of $P(EMD|collision)$ for urban critical bridges shown in Figs. 3(c and d) based on different shear capacities and PSL. Table 14 lists the computed probability values of $P(ESD|collision)$ for urban typical bridges shown in Figs. 4(c and d) based on different shear capacities and PSL.

Table 11. $P(EMD|collision)$ for critical bridges located in rural highways

Critical C (kN)	Rural interstates and arterials				Rural collectors and locals			
	Posted speed limit (kph)				Posted speed limit (kph)			
	≤ 64	80	97	113	≤ 64	80	97	113
220	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
440	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
660	0.9952	1.0000	1.0000	1.0000	0.9872	1.0000	1.0000	1.0000
880	0.7758	0.9992	1.0000	1.0000	0.3697	0.9983	1.0000	1.0000
1,100	0.7517	0.9606	1.0000	1.0000	0.3016	0.8935	1.0000	1.0000
1,320	0.7514	0.7714	0.9960	1.0000	0.3012	0.3648	0.9872	1.0000
1,540	0.7209	0.7486	0.9363	0.9998	0.2888	0.3022	0.8188	0.9995
1,760	0.4856	0.7484	0.8534	0.9990	0.1964	0.3013	0.5900	0.9960
1,980	0.1235	0.7470	0.7693	0.9836	0.0500	0.3009	0.3461	0.9513
2,200	0.0048	0.7216	0.7555	0.9591	0.0021	0.2913	0.3107	0.8858
2,420	0.0000	0.5624	0.7514	0.9188	0.0000	0.2281	0.2987	0.7692
2,640	0.0000	0.2546	0.7487	0.8663	0.0000	0.1027	0.2963	0.6144
2,860	0.0000	0.0200	0.7100	0.8156	0.0000	0.0087	0.2801	0.4727
3,080	0.0000	0.0004	0.5119	0.7525	0.0000	0.0002	0.2019	0.2934
3,300	0.0000	0.0000	0.1215	0.7317	0.0000	0.0000	0.0492	0.2856
3,520	0.0000	0.0000	0.0024	0.4984	0.0000	0.0000	0.0009	0.1929
3,740	0.0000	0.0000	0.0001	0.3064	0.0000	0.0000	0.0000	0.1161
3,960	0.0000	0.0000	0.0000	0.1430	0.0000	0.0000	0.0000	0.0529
4,180	0.0000	0.0000	0.0000	0.0513	0.0000	0.0000	0.0000	0.0192
4,400	0.0000	0.0000	0.0000	0.0136	0.0000	0.0000	0.0000	0.0050
4,620	0.0000	0.0000	0.0000	0.0034	0.0000	0.0000	0.0000	0.0013
4,840	0.0000	0.0000	0.0000	0.0008	0.0000	0.0000	0.0000	0.0003
5,060	0.0000	0.0000	0.0000	0.0001	0.0000	0.0000	0.0000	0.0000
5,280	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
5,500	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table 12. $P(\text{ESD}|\text{collision})$ for typical bridges located in rural highways

Typical C (kN)	Rural interstates and arterials				Rural collectors and locals			
	Posted speed limit (kph)				Posted speed limit (kph)			
	≤ 64	80	97	113	≤ 64	80	97	113
220	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
440	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
660	0.8006	0.9998	1.0000	1.0000	0.4410	0.9996	1.0000	1.0000
880	0.7516	0.7786	0.9973	1.0000	0.3012	0.3845	0.9910	1.0000
1,100	0.7516	0.7495	0.9574	0.9999	0.3012	0.3036	0.8796	0.9997
1,320	0.7305	0.7484	0.8714	0.9993	0.2924	0.3013	0.6396	0.9974
1,540	0.0132	0.7368	0.7810	0.9900	0.0055	0.2969	0.3797	0.9698
1,760	0.0000	0.5624	0.7520	0.9404	0.0000	0.2281	0.3023	0.8287
1,980	0.0000	0.1183	0.7439	0.7951	0.0000	0.0486	0.2939	0.4139
2,200	0.0000	0.0013	0.5840	0.7607	0.0000	0.0006	0.2305	0.3170
2,420	0.0000	0.0000	0.2004	0.7435	0.0000	0.0000	0.0803	0.2916
2,640	0.0000	0.0000	0.0365	0.6866	0.0000	0.0000	0.0161	0.2664
2,860	0.0000	0.0000	0.0034	0.5273	0.0000	0.0000	0.0013	0.2038
3,080	0.0000	0.0000	0.0001	0.2940	0.0000	0.0000	0.0000	0.1111
3,300	0.0000	0.0000	0.0000	0.1114	0.0000	0.0000	0.0000	0.0409
3,520	0.0000	0.0000	0.0000	0.0276	0.0000	0.0000	0.0000	0.0097
3,740	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
3,960	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4,180	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4,400	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4,620	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4,840	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
5,060	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
5,280	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
5,500	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table 13. $P(\text{EMD}|\text{collision})$ for critical bridges located in urban highways

Critical C (kN)	Urban interstates and arterials				Urban collectors and locals			
	Posted speed limit (kph)				Posted speed limit (kph)			
	≤ 64	80	97	113	≤ 64	80	97	113
220	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
440	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
660	0.9931	1.0000	1.0000	1.0000	0.9852	1.0000	1.0000	1.0000
880	0.6877	0.9992	1.0000	1.0000	0.2826	0.9981	1.0000	1.0000
1,100	0.6528	0.9481	1.0000	1.0000	0.1988	0.8776	0.9999	1.0000
1,320	0.6523	0.6809	0.9924	1.0000	0.1984	0.2716	0.9846	1.0000
1,540	0.6254	0.6502	0.9085	0.9998	0.1903	0.2019	0.7886	0.9996
1,760	0.4182	0.6500	0.7954	0.9982	0.1278	0.2005	0.5327	0.9965
1,980	0.1096	0.6494	0.6750	0.9772	0.0320	0.2004	0.2571	0.9472
2,200	0.0036	0.6286	0.6558	0.9435	0.0011	0.1935	0.2154	0.8730
2,420	0.0001	0.4895	0.6491	0.8878	0.0000	0.1504	0.2021	0.7394
2,640	0.0000	0.2169	0.6466	0.8114	0.0000	0.0674	0.1993	0.5671
2,860	0.0000	0.0182	0.6159	0.7422	0.0000	0.0055	0.1884	0.4060
3,080	0.0000	0.0003	0.4421	0.6544	0.0000	0.0001	0.1356	0.1990
3,300	0.0000	0.0000	0.1057	0.6374	0.0000	0.0000	0.0334	0.1930
3,520	0.0000	0.0000	0.0022	0.4337	0.0000	0.0000	0.0006	0.1301
3,740	0.0000	0.0000	0.0002	0.2653	0.0000	0.0000	0.0000	0.0777
3,960	0.0000	0.0000	0.0000	0.1261	0.0000	0.0000	0.0000	0.0374
4,180	0.0000	0.0000	0.0000	0.0431	0.0000	0.0000	0.0000	0.0126
4,400	0.0000	0.0000	0.0000	0.0116	0.0000	0.0000	0.0000	0.0030
4,620	0.0000	0.0000	0.0000	0.0025	0.0000	0.0000	0.0000	0.0007
4,840	0.0000	0.0000	0.0000	0.0006	0.0000	0.0000	0.0000	0.0001
5,060	0.0000	0.0000	0.0000	0.0002	0.0000	0.0000	0.0000	0.0000
5,280	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
5,500	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table 14. $P(\text{ESD}|\text{collision})$ for typical bridges located in urban highways

Typical C (kN)	Urban interstates and arterials				Urban collectors and locals			
	Posted speed limit (kph)				Posted speed limit (kph)			
	≤ 64	80	97	113	≤ 64	80	97	113
220	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
440	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
660	0.7235	0.9997	1.0000	1.0000	0.3667	0.9994	1.0000	1.0000
880	0.6527	0.6917	0.9944	1.0000	0.1984	0.2934	0.9892	1.0000
1,100	0.6527	0.6507	0.9382	0.9999	0.1984	0.2034	0.8609	0.9999
1,320	0.6339	0.6501	0.8206	0.9990	0.1924	0.2006	0.5906	0.9978
1,540	0.0106	0.6402	0.6919	0.9862	0.0036	0.1974	0.2956	0.9679
1,760	0.0001	0.4895	0.6510	0.9158	0.0000	0.1504	0.2062	0.8087
1,980	0.0000	0.1010	0.6432	0.7137	0.0000	0.0314	0.1975	0.3368
2,200	0.0000	0.0008	0.5050	0.6659	0.0000	0.0002	0.1538	0.2252
2,420	0.0000	0.0000	0.1750	0.6480	0.0000	0.0000	0.0550	0.1982
2,640	0.0000	0.0000	0.0338	0.5957	0.0000	0.0000	0.0098	0.1808
2,860	0.0000	0.0000	0.0032	0.4583	0.0000	0.0000	0.0008	0.1369
3,080	0.0000	0.0000	0.0001	0.2550	0.0000	0.0000	0.0000	0.0747
3,300	0.0000	0.0000	0.0000	0.0974	0.0000	0.0000	0.0000	0.0289
3,520	0.0000	0.0000	0.0000	0.0237	0.0000	0.0000	0.0000	0.0065
3,740	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
3,960	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4,180	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4,400	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4,620	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4,840	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
5,060	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
5,280	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
5,500	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

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Data Availability Statement

All data, models, and code generated or used during the study appear in the published article.

References

- AASHTO. 2017. *AASHTO LRFD bridge design specifications*. 8th ed. Washington, DC: AASHTO.
- Agrawal, A. K., S. El-Tawil, R. Cao, X. Xu, X. Chen, and W. Wong. 2018. *A performance based approach for loading definition of heavy vehicle impact events*. Rep. No. FHWA-HIF-18-062. McLean, VA: FHWA.
- Buth, C. E., W. F. Williams, M. S. Brackin, D. Lord, S. R. Geedipally, and A. Y. Abu-Odeh. 2010. *Analysis of large truck collisions with bridge piers: Phase 1. Report of guidelines for designing bridge piers and abutments for vehicle collisions*. Rep. No. FHWA/TX-10/9-4973-1. College Station, TX: Texas Transportation Institute.
- Cao, R., A. K. Agrawal, S. El-Tawil, and W. Wong. 2020. "Numerical studies on concrete barriers subjected to MASH truck impact." *J. Bridge Eng.* 25 (7), 04020035. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0001570](https://doi.org/10.1061/(ASCE)BE.1943-5592.0001570).
- Cao, R., A. K. Agrawal, S. El-Tawil, X. Xu, and W. Wong. 2019a. "Heavy truck collision with bridge piers: Computational simulation study." *J. Bridge Eng.* 24 (6): 04019052. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0001398](https://doi.org/10.1061/(ASCE)BE.1943-5592.0001398).
- Cao, R., A. K. Agrawal, S. El-Tawil, X. Xu, and W. Wong. 2019b. "Performance-based design framework for bridge piers subjected to truck collision." *J. Bridge Eng.* 24 (7), 04019064. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0001423](https://doi.org/10.1061/(ASCE)BE.1943-5592.0001423).
- Cao, R., S. El-Tawil, A. K. Agrawal, X. Xu, and W. Wong. 2019c. "Behavior and design of bridge piers subjected to heavy truck collision." *J. Bridge Eng.* 24 (7), 04019057. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0001414](https://doi.org/10.1061/(ASCE)BE.1943-5592.0001414).
- Chen, X., A. K. Agrawal, S. El-Tawil, X. Xu, R. Cao, and W. Wong. 2019. "Inelastic behavior of a bridge bent subjected to truck impact: Experimental and computational study." *Eng. Struct.* 199: 109543. <https://doi.org/10.1016/j.engstruct.2019.109543>.
- Deng, L., W. He, Y. Yu, and C. S. Cai. 2018. "Equivalent shear force method for detecting the speed and axles of moving vehicles on bridges." *J. Bridge Eng.* 23 (8): 04018057. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0001278](https://doi.org/10.1061/(ASCE)BE.1943-5592.0001278).
- Deng, L., W. Wang, and Y. Yu. 2016. "State-of-the-art review on the causes and mechanisms of bridge collapse." *J. Perform. Constr. Facil.* 30 (2): 04015005. [https://doi.org/10.1061/\(ASCE\)CF.1943-5509.0000731](https://doi.org/10.1061/(ASCE)CF.1943-5509.0000731).
- Hallquist, J. O. 2006. *LS-DYNA theory manual*. Livermore, CA: Livermore Software Technology Corporation.
- He, W., L. Deng, H. Shi, C. S. Cai, and Y. Yu. 2017. "Novel virtual simply supported beam method for detecting the speed and axles of moving vehicles on bridges." *J. Bridge Eng.* 22 (4): 04016141. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0001019](https://doi.org/10.1061/(ASCE)BE.1943-5592.0001019).
- He, W., T. Ling, E. J. O'Brien, and L. Deng. 2019. "Virtual axle method for bridge weigh-in-motion systems requiring no axle detector." *J. Bridge Eng.* 24 (9): 04019086. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0001474](https://doi.org/10.1061/(ASCE)BE.1943-5592.0001474).
- Liu, G. 2012. "Behavior of bridge piers during vehicular impacts." Ph.D. thesis, Dept. of Civil Engineering, City College of New York, City Univ. of New York.

- Miele, C. R., C. Plaxico, D. Stephens, and S. Simunovic. 2010. *U26: Enhanced finite element analysis crash model of tractor-trailers (Phase C)*. Knoxville, TN: National Transportation Research Center, Inc. Univ. Transportation Center.
- Murray, Y. D., A. Y. Abu-Odeh, and R. P. Bligh. 2007. *Evaluation of LS-DYNA concrete material model 159*. FHWA-HRT-05-063. McLean, VA: FHWA.
- NHTSA (National Highway Traffic Safety Administration). 2014. *Traffic safety facts 2014*. Rep. No. DOT HS 812 279. Washington, DC: NHTSA.
- Ray, M. H., C. E. Carrigan, and C. A. Plaxico. 2018. *Guidelines for shielding bridge piers*. Rep. No. NCHRP 892/Project 12-90. Washington, DC: Transportation Research Board, National Research Council.
- Ray, M. H., C. E. Carrigan, C. A. Plaxico, and T. O. Johnson. 2012. *Engineer's manual: Roadside safety analysis program (RSAP) update*. Canton, ME: Roadsafe LLC. Accessed November 22, 2019. <http://rsap.roadsafellc.com/RSAPv3EngineersManual.pdf>.
- Ray, M. H., C. A. Plaxico, and C. E. Carrigan. 2016. *RSAPv3, release 160909BP: Originally documented in roadside safety analysis program (RSAP) engineer's manual*. Canton, ME: Roadsafe LLC. Accessed October 25, 2012. rsap.roadsafellc.com.
- Xu, X., R. Cao, S. El-Tawil, A. K. Agrawal, and W. Wong. 2019. "Loading definition and design of bridge piers impacted by medium weight trucks." *J. Bridge Eng.* 24 (6), 04019042. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0001397](https://doi.org/10.1061/(ASCE)BE.1943-5592.0001397).
- Yi, Z. 2009. "Blast load effects on highway bridges." Ph.D. thesis, Dept. of Civil Engineering, City College of New York, City Univ. of New York.
- Yi, Z., A. K. Agrawal, M. Ettouney, and S. Alampalli. 2013a. "Blast load effects on highway bridges. I: Modeling and blast load effects." *J. Bridge Eng.* 19 (4): 04013023. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0000547](https://doi.org/10.1061/(ASCE)BE.1943-5592.0000547).
- Yi, Z., A. K. Agrawal, M. Ettouney, and S. Alampalli. 2013b. "Blast load effects on highway bridges. II: Failure modes and multi-hazard correlations." *J. Bridge Eng.* 19 (4): 04013024. [https://doi.org/10.1061/\(ASCE\)BE.1943-5592.0000548](https://doi.org/10.1061/(ASCE)BE.1943-5592.0000548).